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6.4.1 Numerical stability of triangular solve

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that we

can expand our argument to yet another element?" And then an inductive argument

would allow us to show it for the entire linear system, for any matrix L that's n by n . Alright, that's the approach. Now what goes into computing χ_1 ? Well,

in exact arithmetic χ_1 would be computed as ψ_1 minus L_{10} transpose times x_0 . And then the result of that would be divided by λ_{11} .

Video

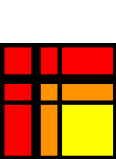
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Reading assignment

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Read Unit 6.4.1 of the notes. [\[LINK\]](#)

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Homework 6.4.1.1

1/1 point (graded)

1/1 point (graded)
Prove Lemma 6.4.1.2.

Hint: Reason the cases where $k = 0$ and $k = 1$ separately from the case where $k > 1$.

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✓ Correct (1/1 point)

Homework 6.4.1.2

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Prove Theorem 6.4.1.3 for the case where $n = 1$.

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Homework 6.4.1.3

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Prove Theorem 6.4.1.3 for the case where $n = 2$.

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Homework 6.4.1.4

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Prove Theorem 6.4.1.3 for the case where $n \geq 1$.

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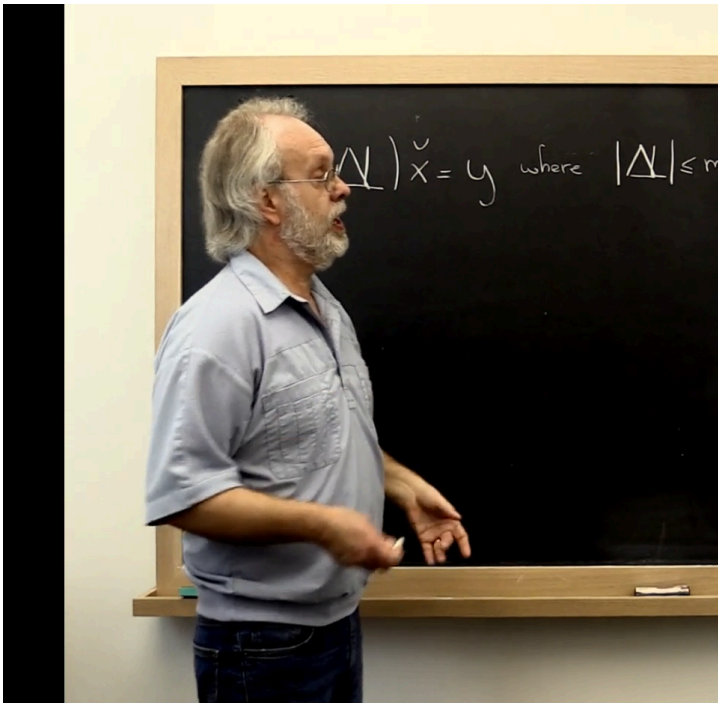
Solution:

For a complete solution, see [the notes](#).

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Answers are displayed within the problem

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error result. Okay the
computed solution to
the lower
triangular system is
the exact solution to
a slightly changed
matrix, a slightly
changed system
where matrix L has
been changed
slightly. And we could
put it
bound on that
element-wise,
because element-
wise is less than or
equal to
the max of gamma_2,
gamma_n-1 times the
absolute value of L.
Okay? Now, since we
**kind of relaxed
bounds all along in
order to come up**



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